Chapter 26

Campbell Biology, AP Edition, 12e

Photograph of a coiled reptile without legs and a serpent-like body.

Chapter 26

Phylogeny and the Tree of Life

Figure 26.1

Although this animal resembles a snake, it is actually a legless lizard: the glass lizard *Ophisaurus apodus*. Why isn't the glass lizard considered a snake? Biologists based this decision in part on a comparison of its traits with those of snakes—and the glass lizard does not have a highly mobile jaw or many vertebrae, two traits shared by all snakes.

How do biologists distinguish and categorize the millions of species on Earth?

Traits shared due to common ancestry are used to classify organisms into groups that reflect their evolutionary history. A tree diagram has a first branch point that represents that common ancestors of all the groups shown here. It splits to fish and another split. The other split is for four limbs. A hatch mark here represents a character shared by the groups to the right of the mark. This branch splits into frogs and then another split. This other split goes through a hatch mark for hair and a branch point. Chimps and humans form a sister group after hair. They are each other's closest relative. The branch point is a common ancestor that had limbs, as do geckos and iguanas, but limbs were lost in the lineages leading to snakes and glass lizards. The branch goes to geckos and then another branch point. That branch point goes to a hatch for no limbs, which leads to snakes. Another direction from the branch goes to a branch for iguanas and a hatch for no limbs. No limbs leads to glass lizards.

Key Concepts

- **26.1Phylogenies show evolutionary relationships****

- **26.2Phylogenies are inferred from morphological and molecular data****
- **26.3Shared characters are used to construct phylogenetic trees****
- **26.4An organism's evolutionary history is documented in its genome****
- **26.5Molecular clocks help track evolutionary time****
- **26.6Our understanding of the tree of life continues to change based on new data****

Advanced placement edition

Phylogenetic trees and cladograms are models of common ancestry and evolutionary history that can be used to establish the relatedness of different species through analysis of similarities in morphology **(BIG IDEA 1)** or comparison of DNA and proteins **(BIG IDEA 3)**.

Study Tip

Label a drawing: The members of a *sister group* are each other's closest relatives, making sister groups a useful way to describe evolutionary relationships. As in this example, identify the sister groups associated with each branch point in Figures 26.11, 26.12, 26.14, and 26.16.

A tree diagram.

The diagram has a split that leads to frogs and another split. It is labeled: frogs form a sister group with the combined chicken, human, mouse lineages. The other split is chicken-human, mouse. It leads to chickens and then to a human, mouse split.

Get Ready for This Chapter

In this chapter, you will learn how biologists use morphological and molecular data to infer phylogeny, the evolutionary history of a species or group of species. Before beginning this chapter, you should be able to do the following:

- Describe evolution as a process of descent with modification (see Concept 22.2, section entitled "Descent with Modification," and Concept 25.2, Figure 25.7).
- Explain what is meant by the genome of an organism (see Concept 12.1 and Concept 21.3).
- Distinguish between homologous and analogous characters (see Concept 22.3).

The following questions provide a quick check of this basic knowledge:

1. Which of the following statements about descent with modification is true?

```
All similarities between organisms result from descent from a common ancestor.
```

That's incorrect. Try again.

Incorrect. Review the following material: All species share at least some characteristics due to the common origin of life. Similarities between two species often result from descent with modification when both species descended from a recent common ancestor. However, similarities between organisms also can result from convergent evolution, the independent evolution of similar features in different evolutionary lineages. Descent with modification also helps to explain differences between organisms: Over many generations, species accumulate differences from their ancestors as they adapt (by natural selection) to different environments.

```
Two species descended from a common ancestor that lived in the distant past will not share any characteristics.
```

That's incorrect. Try again.

Incorrect. Review the following material: All species share at least some characteristics due to the common origin of life. Similarities between two species often result from descent with modification when both species descended from a recent common ancestor. However, similarities between organisms also can result from convergent evolution, the independent evolution of similar features in different evolutionary lineages. Descent with modification also helps to explain differences between organisms: Over many generations, species accumulate differences from their ancestors as they adapt (by natural selection) to different environments.

```
Over time, evolution by natural selection in different environments causes organisms to accumulate differences from their ancestors.
```

That's correct.

Correct! If you had trouble with this question, review the following material: All species share at least some characteristics due to the common origin of life. Similarities between two species often result from descent with modification when both species descended from a recent common ancestor. However, similarities between organisms also can result from convergent evolution, the independent evolution of similar features in different evolutionary lineages. Descent with modification also helps to explain differences between

organisms: Over many generations, species accumulate differences from their ancestors as they adapt (by natural selection) to different environments.

```
        Organisms that share a more recent common ancestor are
likely to be more different from one another than are organisms
that share a distant common ancestor.
```

That's incorrect. Try again.

Incorrect. Review the following material: All species share at least some characteristics due to the common origin of life. Similarities between two species often result from descent with modification when both species descended from a recent common ancestor. However, similarities between organisms also can result from convergent evolution, the independent evolution of similar features in different evolutionary lineages. Descent with modification also helps to explain differences between organisms: Over many generations, species accumulate differences from their ancestors as they adapt (by natural selection) to different environments.

2. The genome of an organism

```
     only includes the organism’s genes.
```

That's incorrect. Try again.

Incorrect. Review the following material: An organism’s DNA includes genes (segments that include proteins) as well as noncoding segments. Organisms within a species display some variation within their DNA sequences.

```
        includes all of the organism’s DNA.
```

That's correct.

Correct!If you had trouble with this question, review the following material: An organism’s genome consists of the complete complement of its genes along with all of its noncoding DNA sequences. Organisms within a species display some variation within their DNA sequences.

```
        only includes segments of the organism’s DNA that do not
encode proteins.
```

That's incorrect. Try again.

Incorrect. Review the following material: An organism's DNA includes genes (segments that include proteins) as well as noncoding segments. Organisms within a species display some variation within their DNA sequences.

```
        of one species is identical to the genome of an organism
of a closely related species.
```

That's incorrect. Try again.

Incorrect. Review the following material: An organism's DNA includes genes (segments that include proteins) as well as noncoding segments. Organisms within a species display some variation within their DNA sequences.

3. Suppose that two distantly related species live in similar environments and share a distinctive characteristic that facilitates survival in their environment but is not found in their immediate ancestors. Based on this information, this distinctive feature most likely represents

```
     an analogous character.
```

That's correct.

Correct! If you had trouble with this question, review the following material: Analogous characters arise by convergent evolution and have similar functions, but not common ancestry. In contrast, homologous features have common ancestry, but not necessarily similar functions. The species described here live in similar environments and share a distinctive feature not found in their common ancestor; hence, it is likely that this feature is an analogous character that arose by convergent evolution as both species adapted to their similar environments.

```
        a character that spread from one species to the other by
gene flow.
```

That's incorrect. Try again.

Incorrect. Review the following material: Analogous characters arise by convergent evolution and have similar functions, but not common ancestry. In contrast, homologous features have common ancestry, but not necessarily similar functions. The species described here live in similar environments and share a

distinctive feature not found in their common ancestor; hence, it is likely that this feature is an analogous character that arose by convergent evolution as both species adapted to their similar environments.

```
a homologous character.
```

That's incorrect. Try again.

Incorrect. Review the following material: Analogous characters arise by convergent evolution and have similar functions, but not common ancestry. In contrast, homologous features have common ancestry, but not necessarily similar functions. The species described here live in similar environments and share a distinctive feature not found in their common ancestor; hence, it is likely that this feature is an analogous character that arose by convergent evolution as both species adapted to their similar environments.

```
a character that arose by genetic drift.
```

That's incorrect. Try again.

Incorrect. Review the following material: Analogous characters arise by convergent evolution and have similar functions, but not common ancestry. In contrast, homologous features have common ancestry, but not necessarily similar functions. The species described here live in similar environments and share a distinctive feature not found in their common ancestor; hence, it is likely that this feature is an analogous character that arose by convergent evolution as both species adapted to their similar environments.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 26.1Define phylogeny and explain how phylogenies are used.
- 26.2Explain how phylogenetic trees are inferred from data.
- 26.3Differentiate between derived and ancestral characters and explain how they are used to construct phylogenetic trees.
- 26.4Describe what evidence a genome provides about evolutionary history.
- 26.5Explain what molecular clocks are and how they are used.
- 26.6Describe the possible structure of the tree of all life.

Concept 26.1: Phylogenies show evolutionary relationships

The glass lizard in Figure 26.1 is part of the continuum of life extending from the earliest organisms to the great variety of species alive today. In this unit, we will survey this diversity and describe hypotheses regarding how it evolved.

To set the stage for surveying life's diversity, in this chapter we consider how biologists trace **phylogeny**, the evolutionary history of a species or group of species. A phylogeny of lizards and snakes, for example, indicates that both glass lizards and snakes evolved from lizards with legs—but they evolved from different lineages of legged lizards **(Figure 26.2)**. Thus, it appears that their legless conditions evolved independently. As we'll see, biologists reconstruct phylogenies like that in Figure 26.2 using **systematics**, a discipline focused on classifying organisms and determining their evolutionary relationships. We'll begin by describing how organisms are named.

Figure 26.2

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Convergent evolution of limbless bodies.
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A phylogenetic tree for the evolution of lizards, snakes, geckos and iguanas from an ancestral lizard with limbs.

A phylogeny based on DNA sequence data reveals that a legless body form evolved independently from legged ancestors in the lineages leading to glass lizards and to snakes.

The tree is as follows. Root ancestral lizard, with limbs, leads to node 1. Node 1 splits to branch tip geckos and node 2. Node 2 splits to branch tip snakes, no limbs, and node 3. Node 3 splits to branch tip iguanas and node 4. Node 4 splits to branch tips monitor lizards and glass lizards, no limbs.

Binomial Nomenclature

Common names for organisms—such as monkey, finch, and lilac—convey meaning in casual usage, but they can also cause confusion. Each of these names, for example, refers to more than one species. Moreover, some common names do not accurately reflect the kind of organism they signify. Consider these three

“fishes”: jellyfish (a cnidarian), crayfish (a small, lobsterlike crustacean), and silverfish (an insect). And, of course, a given organism has different names in different languages.

To avoid ambiguity when communicating about their research, biologists refer to organisms by Latin scientific names. The two-part format of the scientific name, commonly called a **binomial**, was instituted in the 18th century by Carolus Linnaeus (see Concept 22.1). The first part of a binomial is the name of the **genus** (plural, *genera*) to which the species belongs. The second part, called the specific epithet, is unique for each species within the genus. An example of a binomial is *Panthera pardus*, the scientific name for the leopard. Notice that the first letter of the genus is capitalized and the entire binomial is italicized. (Newly created scientific names are also “latinized”: You can name an insect you discover after a friend, but you must add a Latin ending.) Many of the more than 11,000 binomials assigned by Linnaeus are still used today, including the optimistic name he gave our own species—*Homo sapiens*, meaning “wise man.”

Hierarchical Classification

In addition to naming species, Linnaeus also grouped them into a hierarchy of increasingly inclusive categories. The first grouping is built into the binomial: Species that appear to be closely related are grouped into the same genus. For example, the leopard (*Panthera pardus*) belongs to a genus that also includes the African lion (*Panthera leo*), the tiger (*Panthera tigris*), and the jaguar (*Panthera onca*). Beyond genera, biologists employ progressively more comprehensive categories of classification. The classification system named after Linnaeus, the Linnaean system, places related genera in the same **family**, families into **orders**, orders into **classes**, classes into **phyla** (singular, *phylum*), phyla into **kingdoms**, and, more recently, kingdoms into **domains (Figure 26.3)**. The resulting biological classification of a particular organism is somewhat like a postal address identifying a person in a particular apartment, in a building with many apartments, on a street with many apartment buildings, in a city with many streets, and so on.

Figure 26.3

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Linnaean classification.
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Diagram depicts Linnaean classification as series of stacked funnels, from species at top of the stack to domain at bottom. Each level is more inclusive than the one below.

At each level, or “rank,” species are placed in groups within more inclusive groups.

Diagram depicts Linnaean classification as series of stacked funnels, from species at top of the stack to domain at bottom. Each level is more inclusive than the one below. Diagram described top to bottom, with

funnels in the stack numbered 1(top) to 8 (bottom):

1. Drawing of a leopard sits at the top of the stack, on the species funnel. Species is: Panthera pardus.
2. Genus: Panthera
3. Family: Felidae
4. Order: Carnivora
5. Class: Mammalia
6. Phylum: Chordata
7. Kingdom: Animalia
8. Domain: Eukarya.

Two additional categories for Domain are shown surrounding the bottom level (for which the leopard is not a part): Domain bacteria and Domain Archaea.

The named group at any level of the hierarchy is called a **taxon** (plural, *taxa*). In the leopard example, *Panthera* is a taxon at the genus level, and Mammalia is a taxon at the class level that includes all the many orders of mammals. Note that in the Linnaean system, taxa broader than the genus are not italicized, though they are capitalized.

Classifying species is a way to structure our human view of the world. We lump together various species of trees to which we give the common name of pines and distinguish them from other trees that we call firs. Systematists have decided that pines and firs are different enough to be placed in separate genera, yet similar enough to be grouped into the same family, Pinaceae. As with pines and firs, higher levels of classification are usually defined by particular characters chosen by systematists. However, characters that are useful for classifying one group of organisms may not be appropriate for other organisms. For this reason, the larger categories often are not comparable between lineages; that is, an order of snails does not exhibit the same degree of morphological or genetic diversity as an order of mammals. As we'll see, the placement of species into orders, classes, and so on also does not necessarily reflect evolutionary history.

Linking Classification and Phylogeny

The evolutionary history of a group of organisms can be represented in a branching diagram called a **phylogenetic tree**. As in **Figure 26.4**, the branching pattern often matches how systematists have classified groups of organisms nested within more inclusive groups. Sometimes, however, systematists have

placed a species within a genus (or other group) to which it is *not* most closely related. One reason for such a mistake might be that over the course of evolution, a species has lost a key feature shared by its close relatives. If DNA or other new evidence indicates that an organism has been misclassified, the organism may be reclassified to accurately reflect its evolutionary history. Another issue is that while the Linnaean system may distinguish groups, such as amphibians, mammals, reptiles, and other classes of vertebrates, it tells us nothing about these groups' evolutionary relationships to one another.

Figure 26.4

The connection between classification and phylogeny.

Tree diagram depicting the relationship between classification and phylogeny. Order, Family, Genus and Species are labeled on the tree. Branch points represent common ancestors.

Hierarchical classification can reflect the branching patterns of phylogenetic trees. This tree shows evolutionary relationships between some of the taxa within order Carnivora, itself a branch of class Mammalia.

The tree starts at root, order, splits to family and genus branches, then ends with branch tips, species. The tree is as follows. Root, order carnivora, leads to node 1. Node 1 splits to branch family Felidae genus Panthera, species Panthera pardus leopard, and node 2. Node 2 splits to branch family Mustelidae and node 3, and branch family Canidae genus Canis and node 4. Node 3 splits to branch genus Taxidea species Taxidea taxus American badger and branch genus Lutra species Lutra lutra European otter. Node 4 splits to branch species Canis latrans coyote and branch species Canis lupus gray wolf.

Such difficulties in aligning Linnaean classification with phylogeny have led many systematists to propose that classification be based entirely on evolutionary relationships. In such systems, names are assigned only to groups that include a common ancestor and all of its descendants. Using this approach, some commonly recognized groups become part of other groups previously at the same level of the Linnaean system. For example, because birds evolved from a group of reptiles, Aves (the Linnaean class to which birds are assigned) is considered a subgroup of Reptilia (also a class in the Linnaean system).

What We Can and Cannot Learn from Phylogenetic Trees

Regardless of how groups are named, a phylogenetic tree represents a hypothesis about evolutionary relationships. These relationships often are depicted as a series of dichotomies, or two-way branch points. Each **branch point** represents the common ancestor of the two evolutionary lineages diverging from it. An **evolutionary lineage** is a sequence of ancestral organisms leading to a particular descendant taxon.

Figure 26.5 illustrates these elements of a phylogenetic tree and offers more tips on interpreting these diagrams.

Advanced placement edition

Apply concepts and connect across Big Ideas using visual representations.

What are some of the assumptions this model makes about derived or lost characteristics at each branch point? Where does *Dimetrodon*, an extinct carnivore that was more closely related to mammals than to reptiles, fit on the tree?

Figure 26.5

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Visualizing Phylogenetic Relationships
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Tree diagram and two alternative forms of tree diagrams with instructions on how to read the diagrams.

A phylogenetic tree visually represents a hypothesis of how a group of organisms are related. This figure explores how the way a tree is drawn conveys information.

Parts of a tree: This tree shows how the five groups of organisms at the tips of the branches, called taxa, are related. Each branch point represents the common ancestor of the evolutionary lineages diverging from it. Tree is described as a nested list with the top of the list corresponding to the far left of the diagram and moving right as the list continues. All labels and instructions are noted where given.

Farthest left branch point: This branch point represents the common ancestor of all the animal groups shown in this tree. At the far right of the diagram, five taxa are listed. Extending to the right, each horizontal branch represents an evolutionary lineage. The length of the branch is arbitrary unless the diagram specifies that branch lengths represent information such as time or amount of genetic change (see Figure 26.13). Each position along a branch represents an ancestor in the lineage leading to the taxon named at the tip. Two arms extend to the right of the common ancestor. The top arm leads to fishes. The bottom arm branches into two more arms. The top branch leads to frogs. The bottom branch leads to two more branches. The top branch leads to lizards. The bottom branch leads to two more branches. The top branch leads to chimps. The bottom branch leads to humans. Sister taxa are groups of organisms that share a common ancestor that is not shared by any other group. Chimps and humans are an example of sister taxa in this tree.

Alternative forms of tree diagrams: These diagrams are referred to as trees because they use the visual analogy of branches to represent evolutionary lineages diverging over time. In this text, trees are usually drawn horizontally, as shown above, but the same tree can be drawn vertically or diagonally without changing the relationships it conveys. There are two tree diagrams that show the same five taxa as shown

above: fishes, frogs, lizards, chimps, and humans. In the diagram on the left (a vertical tree), the common ancestor is shown at the bottom and the arms of the tree branch vertically upward, with the same interrelationships between the taxa as shown above. In the diagram on the right (a diagonal tree), the common ancestor is shown at bottom left and the lineage line runs diagonally to the top right. Each taxon branches diagonally to the top left from this line. In sequence from left to right, the branches are labeled fishes, frogs, lizards, and chimps. Humans are labeled at top right at the end point of the diagonal tree.

Rotating around branch lengths: Rotating the branches of a tree around a branch point does not change what they convey about evolutionary relationships. As a result, the order in which taxa appear at the branch tips is not significant. What matters is the branching pattern, which signifies the order in which the lineages have diverged from common ancestors. This section shows two diagrams. The diagram on the left repeats the horizontal tree with five taxa from the beginning of this figure. Rotating the branches in the tree at left around the three blue branch points yields the tree on the right. In the tree on the right, the common ancestor is now at the bottom left and the bottom arm now leads to fishes. However, the interrelationships between the taxa are still the same. Note: The order of the taxa does NOT represent a sequence of evolution leading to the last taxon shown.

Instructors: Additional questions related to this Visualizing Figure can be assigned in **Mastering Biology**.

Each tree in Figure 26.5 has a branch point that represents the common ancestor of chimpanzees and humans. Chimps and humans are an example of **sister taxa**, groups of organisms that share an immediate common ancestor that is not shared by any other group. Note that there is a sister group associated with each branch point in a tree (since each branch point represents the common ancestor of the two lineages diverging at that point).

The members of a sister group are each other's closest relatives, making sister groups a useful way to describe the evolutionary relationships shown in a tree. For example, in Figure 26.5, the evolutionary lineage leading to lizards shares an immediate common ancestor with the lineage leading to chimpanzees and humans. Thus, we can describe this portion of the tree by saying that lizards are the sister taxon to a group consisting of chimpanzees and humans.

As also discussed in Figure 26.5, the branches of a tree can be rotated around branch points without changing the relationships shown in the tree. That is, the order in which the taxa appear at the right side of the tree does not represent a *sequence* of evolution—in this case, it does not imply a sequence leading from fishes to humans.

This tree, like all of the phylogenetic trees in this book, is **rooted**, which means that a branch point within the tree (often drawn farthest to the left) represents the most recent common ancestor of all taxa in the tree.

A lineage that diverges from all other members of its group early in the history of the group is called a **basal taxon**. Hence, like the fishes in Figure 26.5, a basal taxon lies on one of the two branches diverging from the branch point that represents the common ancestor of the group. In general, the lineages represented by these two branches originated at the same point in time and hence have been evolving for the same length of time.

What other key points do we need to keep in mind when interpreting phylogenetic trees? First, they are intended to show patterns of descent, not phenotypic similarity. Although closely related organisms often resemble one another due to their common ancestry, they may not if their lineages have evolved at different rates or faced very different environmental conditions. For example, even though crocodiles are more closely related to birds than to lizards (see Figure 22.17), they look more like lizards because morphology has changed dramatically in the bird lineage.

Second, we cannot necessarily infer the ages of the taxa or branch points shown in a tree. For example, the tree in Figure 26.5 does not indicate that chimpanzees evolved before humans. Rather, the tree shows only that chimpanzees and humans share a recent common ancestor, but we cannot tell when that ancestor lived or when the first chimpanzees or humans arose. Generally, unless we are given specific information about what the branch lengths in a tree mean—for example, that they are proportional to time—we should interpret the diagram solely in terms of patterns of descent. No assumptions should be made about when particular species evolved or how much change occurred in each lineage.

Third, we should not assume that a taxon on a phylogenetic tree evolved from the taxon next to it. Figure 26.5 does not indicate that humans evolved from chimpanzees or vice versa. We can infer only that the lineage leading to humans and the lineage leading to chimpanzees both evolved from a recent common ancestor. That ancestor, which is now extinct, was neither a human nor a chimpanzee.

CheckPoint

Reading Phylogenetic Trees

Check your understanding before you move on, with the CheckPoint: Reading Phylogenetic Trees.

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Applying Phylogenies

Understanding phylogeny can have practical applications. Consider maize (corn), which originated in the Americas and is now an important food crop worldwide. From a phylogeny of maize based on DNA data,

researchers have identified two species of wild grasses that may be maize's closest living relatives. These two close relatives may be useful as "reservoirs" of beneficial alleles that can be transferred to cultivated maize by cross-breeding or genetic engineering.

A different use of phylogenetic trees is to infer species identities by analyzing the relatedness of DNA sequences from different organisms. Researchers have used this approach to investigate whether "whale meat" had been harvested illegally from whale species protected under international law rather than from species that can be harvested legally **(Figure 26.6)**.

Figure 26.6

Inquiry: What is the species identity of food being sold as whale meat?

Experiment

C. S. Baker and S. R. Palumbi purchased 13 samples of "whale meat" from Japanese fish markets. They sequenced part of the mitochondrial DNA (mtDNA) from each sample and compared their results with the comparable mtDNA sequence from known whale species. To infer the species identity of each sample, the team constructed a *gene tree*, a phylogenetic tree that shows patterns of relatedness among DNA sequences rather than among taxa.

Results

Of the species in the resulting gene tree, only Minke whales caught in the Southern Hemisphere can be sold legally in Japan.

A phylogenetic tree depicting patterns of relatedness among DNA sequences from whale meat.

The tree is as follows. Root leads to node 1. Node 1 splits to nodes 2 and 3. Node 2 splits to nodes 4 and 5. Node 4 splits to branch tip Minke, southern hemisphere, m t D N A, and branch tip unknown m t D N A number 1 a, 2, 3, 4, 5, 6, 7, 8. Node 5 splits to branch tip Minke, north Atlantic, m t D N A, and branch tip unknown m t D N A number 9. Node 3 splits to nodes 6 and 7. Node 6 splits to node 8 and branch tip blue m t D N A. Node 8 splits to branch tip humpback m t D N A and unknown m t D N A number 1 b. Node 7 splits to branch tip unknown m t D N A number 10, 11, 12, 13, and branch tip fin m t D N A.

Conclusion

This analysis indicated that mtDNA sequences of six of the unknown samples (in red) were most closely related to mtDNA sequences of whales that are not legal to harvest.

Data from C. S. Baker and S. R. Palumbi, Which whales are hunted? A molecular genetic approach to monitoring whaling, *Science* 265:1538–1539 (1994). Reprinted with permission from AAAS.

WHAT IF?*What different results would have indicated that all of the whale meat had been harvested legally?*

*For suggested answer, see*Appendix A.**

How do researchers construct phylogenetic trees like those we've considered here? In the next section, we'll begin to answer that question by examining the data used to determine phylogenies.

Concept 26.2: Phylogenies are inferred from morphological and molecular data

To infer phylogeny, systematists must gather as much information as possible about the morphology, genes, and biochemistry of the relevant organisms. It is important to focus on features that result from common ancestry because only those features reflect evolutionary relationships.

Morphological and Molecular Homologies

Recall that phenotypic and genetic similarities due to shared ancestry are called **homologies**. For example, the similarity in the number and arrangement of bones in the forelimbs of mammals is due to their descent from a common ancestor with the same bone structure; this is an example of a morphological homology (see Figure 22.15). In the same way, genes or other DNA sequences are homologous if they are descended from sequences carried by a common ancestor.

In general, organisms that share very similar morphologies or similar DNA sequences are likely to be more closely related than organisms with vastly different structures or sequences. In some cases, however, the morphological divergence between related species can be great and their genetic divergence small (or vice versa). Consider the Hawaiian silversword plants: some of these species are tall, twiggy trees, while others are dense, ground-hugging shrubs (see Figure 25.23). But despite these striking phenotypic differences, the

silverswords' genes are very similar. Based on these small molecular divergences, scientists estimate that the silversword group began to diverge 5 million years ago. We'll discuss how scientists use molecular data to estimate such divergence times later in this chapter.

Sorting Homology from Analogy

A potential source of confusion in constructing a phylogeny is similarity between organisms that is due to convergent evolution—called **analogy**—rather than to shared ancestry (homology). Convergent evolution occurs when similar environmental pressures and natural selection produce similar (analogous) adaptations in organisms from different evolutionary lineages. For example, the two mole-like animals shown in **Figure 26.7** look very similar. However, their internal anatomy, physiology, and reproductive systems are very dissimilar. Indeed, genetic and fossil evidence indicate that the common ancestor of these animals lived 160 million years ago. This common ancestor and most of its descendants were not mole-like. It appears that analogous characteristics evolved independently in these two lineages as they became adapted to similar lifestyles—hence, the similar mole-like features of these animals should not be considered when reconstructing their phylogeny.

Figure 26.7

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Convergent evolution in burrowers.
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Drawings of an Australian marsupial mole and a North American eutherian mole. Appearance is similar but differences in facial and foot structure are apparent.

A long body, large front paws, small eyes, and a pad of thick skin that protects the nose all evolved independently in these species.

Another clue to distinguishing between homology and analogy is the complexity of the characters being compared. The more elements that are similar in two complex structures, the more likely it is that the structures evolved from a common ancestor. For instance, the skulls of an adult human and an adult chimpanzee both consist of many bones fused together. The compositions of the skulls match almost perfectly, bone for bone. It is highly improbable that such complex structures, matching in so many details, have separate origins. More likely, the genes involved in the development of both skulls were inherited from a common ancestor.

The same argument applies to comparisons at the gene level. Genes are sequences of thousands of nucleotides, each of which represents an inherited character in the form of one of the four DNA bases: A

(adenine), G (guanine), C (cytosine), or T (thymine). If genes in two organisms share many portions of their nucleotide sequences, it is likely that the genes are homologous.

Evaluating Molecular Homologies

Comparing DNA molecules often poses technical challenges for researchers. The first step after sequencing the DNA is to align comparable sequences from the species being studied. If the species are very closely related, the sequences probably differ at only one or a few sites. In contrast, comparable nucleic acid sequences in distantly related species usually have different bases at many sites and may have different lengths. This is because insertions and deletions accumulate over long periods of time.

Suppose, for example, that certain noncoding DNA sequences near a particular gene are very similar in two species, except that the first base of the sequence has been deleted in one of the species. The effect is that the remaining sequence shifts back one notch. A comparison of the two sequences that does not take this deletion into account would overlook what in fact is a very good match. Computer programs can be used to identify such matches by testing possible alignments for comparable DNA segments of differing lengths **(Figure 26.8)**.

Figure 26.8

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Aligning segments of DNA.
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Aligning segments of D N A.

Steps are listed as follows.

1. Homologous DNA sequences are identical as species 1 and species 2 begin to diverge from their common ancestor. Sequence 1. C,C, A, T, C, A, G, A, G, T, C. Sequence 2. C, C, A, T, C, A, G, A, G, T, C.
2. Deletion and insertion mutations shift what had been matching sequences in the two species. In Sequence 1, G is deleted and in Sequence 2, G T A is inserted between T and C.
3. Of the regions of the species 2 sequence that match the species 1 sequence, only the first 4 sequences (C C A T) align because of these mutations. Remaining regions no longer align because of these mutations.
4. The matching regions realign after a computer program adds gaps in sequence 1 where G T A and G occur in sequence 2.

Systematists search for similar sequences along DNA segments from two species (only one DNA strand is shown for each species). In this example, 11 of the original 12 bases have not changed since the species diverged. Hence, those portions of the sequences still align once the length is adjusted.

Such molecular comparisons reveal that many base substitutions and other differences have accumulated in the comparable genes of an Australian “mole” and a golden mole. The many differences indicate that their lineages have diverged greatly since their common ancestor; thus, we say that the living species are not closely related. In contrast, the high degree of gene sequence similarity among the silversword plants indicates that they are all very closely related, in spite of their considerable morphological differences.

Just as with morphological characters, it is necessary to distinguish homology from analogy in evaluating molecular similarities for evolutionary studies. Two sequences that resemble each other at many points along their length most likely are homologous (see Figure 26.8). But in organisms that do not appear to be closely related, the bases that their otherwise very different sequences happen to share may simply be coincidental matches **(Figure 26.9)**. Scientists have developed statistical tools that can help distinguish “distant” homologies from such coincidental matches in extremely divergent sequences.

Figure 26.9

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A coincidental match.
```

Two D N A sequences each have 22 bases. 5 bases are aligned.

Top sequence: A C G G A T A G T C C A C T A G G C A C T A. Bottom sequence: T C A C C G A C A G G T C T T T G A C T A G. Aligned bases in both the sequences are highlighted in yellow: C (at the second position), A (at seventh position), C T (at the thirteenth and fourteenth positions), and G (at the seventeenth position).

These two DNA sequences from distantly related species coincidentally share 23% of their bases (those highlighted in yellow).

WHAT IF?**Why might you expect organisms that are not closely related to nevertheless share roughly*25%*of their bases?*

*For suggested answer, see**Appendix A*.*

Concept 26.3: Shared characters are used to construct phylogenetic trees

As we've discussed, a key step in reconstructing phylogenies is to distinguish homologous features from analogous ones (since only homology reflects evolutionary history). Next, we'll describe cladistics, a widely used set of methods for inferring phylogeny.

Cladistics

In the approach to systematics called **cladistics**, common ancestry is the primary criterion used to classify organisms. Using this methodology, biologists attempt to place species into groups called **clades**, each of which includes an ancestral species and all of its descendants. Clades, like the categories of the Linnaean system, are nested within larger clades. In Figure 26.4, for example, the cat group (Felidae) represents a clade within a larger clade (Carnivora) that also includes the dog group (Canidae).

A taxon is equivalent to a clade only if it is **monophyletic** (from the Greek, meaning "single tribe"), signifying that it consists of an ancestral species and all of its descendants **(Figure 26.10a)**. Contrast this with a **paraphyletic** ("beside the tribe") group, which consists of an ancestral species and some, but not all, of its descendants **(Figure 26.10b)**, or a **polyphyletic** ("many tribes") group, which includes distantly related species but does not include their most recent common ancestor **(Figure 26.10c)**.

Figure 26.10

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Monophyletic, paraphyletic, and polyphyletic groups.
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A phylogenetic tree illustrates three different groups.

The tree is as follows. Root is leads to nodes 1. Node 1 is labeled common ancestor 3. It splits to nodes 2 and 3. Node 2 is labeled common ancestor 1. It splits to branch tip A and node 4. Node 4 splits to branch tips B and C. Node 3 is labeled common ancestor 2. It splits to nodes 5 and 6. Node 5 splits to branch tips D and E. Node 6 splits to branch tips F and G. Common ancestor 1 and branches A, B and C are highlighted, representing group 1. Group 1, consisting of three species (A, B, C) and their common ancestor 1, is a monophyletic group (clade), meaning that it consists of an ancestral species and all of its descendants. Common ancestor 2 and branches D, E, and F are highlighted, representing group 2. Group 2 is paraphyletic, meaning that it consists of an ancestral species 2 and some of its descendants (species D, E, F)

but not all of them (does not include species G). Branches A, B, C, and D are highlighted, representing group 3. Group 3, consisting of four species (A, B, C, D), is polyphyletic, meaning that the most recent common ancestor, 3, of its members is not part of the group.

Note that in a paraphyletic group, the most recent common ancestor of all members of the group *is* part of the group, whereas in a polyphyletic group the most recent common ancestor *is not* part of the group. For example, a group consisting of even-toed ungulates (hippopotamuses, deer, and their relatives) and their common ancestor is paraphyletic because it does not include cetaceans (whales, dolphins, and porpoises), which descended from that ancestor **(Figure 26.11)**. In contrast, a group consisting of seals and cetaceans (based on their similar body forms) is polyphyletic because it does not include the common ancestor of seals and cetaceans. Biologists avoid defining such polyphyletic groups; if new evidence indicates that an existing group is polyphyletic, its members are reclassified.

Figure 26.11

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Paraphyletic versus polyphyletic groups.
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A phylogenetic tree showing paraphyletic and polyphyletic groups.

The tree is as follows. Root leads to node 1. Node 1 splits to nodes 2 and 3. Node 2 represents common ancestor of even-toed ungulates. It splits to branch tip, other even-toed ungulates, and node 4. Node 4 splits to branch tips hippopotamuses and cetaceans. Node 3 splits to node 5 and branch tip, other carnivores. Node 5 splits to branch tips seals and bears. The paraphyletic group example consists of the other even-toed ungulates and the hippopotamuses, because this grouping does not include all the descendants of the common ancestor (it excludes cetaceans). The polyphyletic group example consists of the cetaceans and seals, because this group does not include the most recent common ancestor of its members.

DRAW IT *Circle the branch point that represents the most recent common ancestor of cetaceans and seals. Explain why that ancestor would not be part of a cetacean–seal group defined by similar body form.*

*For suggested answer, see*Appendix A.**

Shared Ancestral and Shared Derived Characters

As a result of descent with modification, organisms have characters they share with their ancestors, and they also have characters that differ from those of their ancestors. For example, all mammals have backbones, but a backbone does not distinguish mammals from other vertebrates because *all* vertebrates have backbones. The backbone predates the branching of mammals from other vertebrates. Thus for mammals, the backbone is a **shared ancestral character**, a character that originated in an ancestor of the taxon. In contrast, hair

is a character shared by all mammals but *not* found in their ancestors. Thus, in mammals, hair is considered a **shared derived character**, an evolutionary novelty unique to a clade.

Note that a shared derived character can refer to the loss of a feature, such as the loss of limbs in snakes or whales. Also, keep in mind that it is a relative matter whether a character is considered ancestral or derived. A backbone can also qualify as a shared derived character, but only at a deeper branch point that distinguishes all vertebrates from other animals.

Inferring Phylogenies Using Derived Characters

Shared derived characters are unique to particular clades. Because all features of organisms arose at some point in the history of life, it should be possible to determine the clade in which each shared derived character first appeared and to use that information to infer evolutionary relationships.

To give an example of this approach, consider the set of characters shown in **Figure 26.12a** for each of five vertebrates—a leopard, turtle, frog, bass, and lamprey (a jawless aquatic vertebrate). As a basis of comparison, we need to select an outgroup. An **outgroup** is a species or group of species from an evolutionary lineage that is closely related to but not part of the group of species that we are studying (the **ingroup**). A suitable outgroup can be determined based on evidence from morphology, paleontology, embryonic development, and gene sequences. An appropriate outgroup for our example is the lancelet, a small animal that lives in mudflats and (like vertebrates) is a member of the more inclusive group called the chordates. Unlike the vertebrates, however, the lancelet does not have a backbone.

Figure 26.12

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Using derived characters to infer phylogeny.
```

Using derived characters to infer phylogeny.

Part a. Character table. A 0 indicates that the character is absent; a 1 indicates that the character is present.

Part b. Phylogenetic tree for the evolution of vertebrates. The character table in a leads us to infer the phylogenetic relationships shown in the tree, which is as follows. Root leads to node 1. Node 1 splits to branch tip lancelet, outgroup, and node 2 with characteristic vertebral column. Node 2 splits to branch tip lamprey and node 3 with characteristic hinged jaws. Node 3 splits to branch tip bass and node 4 with characteristic four limbs. Node 4 splits to branch tip frog and node 5 with characteristic amnion. Node 5 splits to branch tip turtle and leopard, with characteristic hair.

Characters	Lancelet (outgroup)	Lamprey	Bass	Frog	Turtle	Leopard
Vertebral column (backbone)	0	1	1	1	1	1
Hinged jaws	0	0	1	1	1	1
Four walking legs	0	0	0	1	1	1
Amnion	0	0	0	0	1	1
Hair	0	0	0	0	0	1

The derived characters used here include the amnion, a membrane that encloses the embryo inside a fluid-filled sac (see Figure 34.26). Note that a different set of characters could lead us to infer a different phylogenetic tree.

DRAW IT*In (b), circle the most inclusive clade for which a hinged jaw is a shared* ancestral *character*.

*For suggested answer, see*Appendix A.*

Figure 26.12 Walkthrough

Using derived characters to infer phylogeny

Watch an explanation of this figure in the Figure 26.12 Walkthrough: Using derived characters to infer phylogeny.

image

Play

image

Play

**

In our analysis, a character found in both the outgroup and the ingroup is assumed to be ancestral. We'll also assume that each derived character in Figure 26.12a arose only once in the ingroup. Thus, for a character that only occurs in a subset of the ingroup, we'll assume that the character arose in the lineage leading to those members of the ingroup.

By comparing members of the ingroup with each other and with the outgroup, we can determine which characters were derived at the various branch points of vertebrate evolution. In our example, *all* of the vertebrates in the ingroup have backbones: This character was present in the ancestral vertebrate, but not in the outgroup. Now note that hinged jaws are absent in the outgroup and in lampreys, but present in all other members of the ingroup. This indicates that hinged jaws arose in a lineage leading to all members of the ingroup *except* lampreys. Hence, we can conclude that lampreys are the sister taxon to the other vertebrates in the ingroup. Proceeding in this way, we can translate the data in our table of characters into a phylogenetic tree that places all the ingroup taxa into a hierarchy based on their derived characters **(Figure 26.12b)**.

Phylogenetic Trees with Proportional Branch Lengths

In the phylogenetic trees we have presented so far, the lengths of the tree's branches do not indicate the degree of evolutionary change in each lineage. Furthermore, the chronology represented by the branching pattern of the tree is relative (earlier versus later) rather than absolute (how many millions of years ago). But in some tree diagrams, branch lengths are proportional to the amount of evolutionary change or to the times at which particular events occurred.

In **Figure 26.13**, for example, each branch length of the phylogenetic tree reflects the number of changes that have taken place in a particular DNA sequence in that lineage. Note that the total length of the horizontal lines from the base of the tree to the mouse is less than that of the line leading to the fruit fly, *Drosophila*. This indicates that fewer genetic changes have occurred in the mouse lineage than in the *Drosophila* lineage. In addition, because the mouse and fly lineages have evolved for an equal amount of time after they diverged from a common ancestor, the *rate* of change was lower in mouse lineage than in the *Drosophila* lineage.

Figure 26.13

Branch lengths can represent genetic change.

Phylogenetic tree where branch length represents degree of genetic change.

This tree was based on sequences of homologs of a gene that plays a role in development; Drosophila was used as an outgroup. Branch lengths are proportional to the amount of genetic change in each lineage; varying branch lengths indicate that the gene has evolved at different rates in different lineages.

The tree is as follows, from top to bottom. Note, branches decrease in length from top to bottom. Root leads to node 1. Node 1 splits to branch top drosophila and node 2. Node 2 splits to branch tip lancelet and node 3.

Node 3 splits to branch tip zebrafish and node 4. Node 4 splits to branch tip frog and node 5. Node 5 splits to branch tip chicken and node 6. Node 6 splits to branch tips human and mouse, each the same length.

INTERPRET THE DATA *In which vertebrate lineage has the studied gene evolved most rapidly? Explain.*

In general, although the branches of a phylogenetic tree may have different lengths, among organisms alive today, all the different lineages that descend from a common ancestor have survived for the same number of years. To take an extreme example, humans and bacteria had a common ancestor that lived over 3 billion years ago. Fossils and genetic evidence indicate that this ancestor was a single-celled prokaryote. Even though bacteria have apparently changed little in their morphology since that common ancestor, there have nonetheless been 3 billion years of evolution in the bacterial lineage, just as there have been 3 billion years of evolution in the lineage that ultimately gave rise to humans.

These equal spans of chronological time can be represented in a phylogenetic tree whose branch lengths are proportional to time **(Figure 26.14)**. Such a tree draws on fossil data to place branch points in the context of geologic time. Additionally, it is possible to combine these two types of trees by labeling branch points with information about rates of genetic change or dates of divergence.

Figure 26.14

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Branch lengths can indicate time.
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Evolutionary tree with branch lengths mapped to dates based on fossil evidence. Time scale is millions of years ago, from over 500 million to present.

The previous tree is displayed with branch lengths measured by a timeline as follows.

Drosophila, 620 million years.

Lancelet, 550 million years.

Zebrafish, 410 million years.

Frog, 350 million years.

Chicken, 310 million years.

Human, 60 million years.

Mouse, 60 million years.

All values estimated.

This tree is based on the same DNA data as that in Figure 26.13, but here the branch points are dated based on fossil evidence. Thus, the branch lengths are proportional to time. Each lineage has the same total length from the base of the tree to the branch tip, indicating that all the lineages have diverged from the common ancestor for equal amounts of time.

Maximum Parsimony and Maximum Likelihood

As the database of DNA sequences that enables us to study more species grows, the difficulty of building the phylogenetic tree that best describes their evolutionary history also grows. What if you are analyzing data for 50 species? There are 3×1076 different ways to arrange 50 species into a tree! And which tree in this huge forest reflects the true phylogeny? Systematists can never be sure of finding the most accurate tree in such a large data set, but they can narrow the possibilities by applying the principles of maximum parsimony and maximum likelihood.

According to the principle of **maximum parsimony**, we should first investigate the simplest explanation that is consistent with the facts. (The parsimony principle is also called “Occam’s razor” after William of Occam, a 14th-century English philosopher who advocated solving problems by “shaving away” unnecessary complications.) In the case of trees based on morphology, the most parsimonious tree requires the fewest evolutionary events, as measured by the origin of shared derived morphological characters. For phylogenies based on DNA, the most parsimonious tree requires the fewest base changes.

A **maximum likelihood** approach identifies the tree most likely to have produced a given set of DNA data based on certain probability rules about how DNA sequences change over time. For example, the underlying probability rules could be based on the assumption that all nucleotide substitutions are equally likely. If evidence suggests that this assumption is not correct, more complex rules could be devised to account for different rates of change among different nucleotides or at different positions in a gene.

Scientists have developed many computer programs to search for trees that are parsimonious and likely. When a large amount of accurate data is available, the methods used in these programs usually yield similar trees. As an example of one method, **Figure 26.15** walks you through the process of identifying the most parsimonious molecular tree for a three-species problem. Computer programs use the principle of parsimony to estimate phylogenies in a similar way: They examine large numbers of possible trees and identify those that require the fewest evolutionary changes.

Figure 26.15

Research Method: Applying Parsimony to a Problem in Molecular Systematics

Application

In considering possible phylogenies for a group of species, systematists compare molecular data for the species. An efficient way to begin is by identifying the most parsimonious hypothesis—the one that requires the fewest evolutionary events (molecular changes) to have occurred.

Technique

Follow the numbered steps as we apply the principle of parsimony to a hypothetical phylogenetic problem involving three closely related beetle species.

Applying parsimony to a problem in molecular systematics.

Part 1. Three species of beetles drawn, each with different colors and markings. Part 2. Three possible phylogenetic hypotheses are drawn as simple tree diagrams as follows.

- Hypothesis 1. Root leads to node 1. Node 1 splits to node 2 and branch tip 3. Node 2 splits to branch tips 1 and 2.
- Hypothesis 2. Root leads to node 1. Node 1 splits to node 2 and branch tip 2. Node 2 splits to branch tips 1 and 3.
- Hypothesis 3. Root leads to node 1. Node 1 splits to node 2 and branch tip 1. Node 2 splits to branch tips 3 and 2.

Part 4. The three tree diagrams are updated to include the 1 C base change, noted in parentheses located at the following points.

- Hypothesis 1, node 2.
- Hypothesis 2, branch 1, branch 2.
- Hypothesis 3, branch 2, branch 1.

Part 5. Base change events are noted on each of the three tree diagrams as follows.

- Hypothesis 1. Node 2, 2 T. Branch 3, 3 A and 4 C. Branch 1, 3 A. Branch 2, 4 C.
- Hypothesis 2. Node 2, 3 A. Branch 2, 2 T and 4 C. Branch 1, 2 T. Branch 3, 4 C.

- Hypothesis 3. Node 2, 4 C. Branch 1, 2 T and 3 A. Branch 3, 3 A. Branch 2, 2 T.

Part 6. Base change events are totaled for each of the three trees with hash marks, indicating that Tree 1 is most parsimonious (requires the fewest number of base change events). Events are marked as follows.

- Hypothesis 1. Node 2, 2 events. Branch 3, 2 events. Branch 1, 1 event. Branch 2, 1 even. 6 total events.
- Hypothesis 2. Node 2, 1 event. Branch 2, 3 events. Branch 1, 2 events. Branch 3, 1 event. 7 total events
- Hypothesis 3. Node 2, 1 event. Branch 1, 3 events. Branch 3, 1 event. Branch 2, 2 events. 7 total events.

Part 3. A table is as follows.

blank	Site 1	Site 2	Site 3	Site 4
Species I	C	T	A	T
Species II	C	T	T	C
Species III	A	G	A	C
Ancestral sequence	A	G	T	T

- - STEP 1: First, draw the three possible trees for the species. (Although only three trees are possible when ordering three species, the number of possible trees increases rapidly with the number of species: There are 15 trees for four species and 34,459,425 trees for 10 species.)

- - STEP 2: Tabulate the molecular data for the species. In this simplified example, the data represent a DNA sequence consisting of just four nucleotide bases. Data from several outgroup species (not shown) were used to infer the ancestral DNA sequence.

- - STEP 3: Now focus on site 1 in the DNA sequence. In the tree on the left, a single base-change event, represented by the purple hatch mark on the branch leading to species I and II (and labeled 1/C, indicating a change at site 1 to nucleotide C), is sufficient to account for the site 1 data. In the other two trees, two base-change events are necessary.

- - STEP 4: Continuing the comparison of bases at sites 2, 3, and 4 reveals that each of the

three trees requires a total of five additional base-change events (purple hatch marks).

Results

To identify the most parsimonious tree, we total all of the base-change events noted in steps 3 and 4. We conclude that the first tree is the most parsimonious of the three possible phylogenies. (In a real example, many more sites would be analyzed. Hence, the trees would often differ by more than one base-change event.)

Phylogenetic Trees as Hypotheses

This is a good place to reiterate that any phylogenetic tree represents a hypothesis about how the organisms in the tree are related to one another. The best hypothesis is the one that best fits all the available data. A phylogenetic hypothesis may be modified when new evidence compels systematists to revise their trees. Indeed, while many older phylogenetic hypotheses have been supported by new morphological and molecular data, others have been changed or rejected.

Thinking of phylogenies as hypotheses also allows us to use them in a powerful way: We can make and test predictions based on the assumption that a particular phylogeny—our hypothesis—is correct. For example, in an approach known as *phylogenetic bracketing*, we can predict (by parsimony) that features shared by two groups of closely related organisms are present in their common ancestor and all of its descendants unless independent data indicate otherwise. (Note that “prediction” can refer to unknown past events as well as to evolutionary changes yet to occur.)

This approach has been used to make novel predictions about dinosaurs. For example, there is evidence that birds descended from the theropods, a group of bipedal dinosaurs. As seen in **Figure 26.16**, the closest living relatives of birds are crocodiles. Birds and crocodiles share numerous features: They have four-chambered hearts, they “sing” to defend territories and attract mates (although a crocodile’s “song” is more like a bellow), and they build nests. Both birds and crocodiles also care for their eggs by warming them. Birds warm their eggs by sitting on them, whereas crocodiles cover their eggs with their neck or with sand or vegetation. Reasoning that any feature shared by birds and crocodiles is likely to have been present in their common ancestor (denoted by the blue dot in Figure 26.16) and *all* of its descendants, biologists predicted that dinosaurs had four-chambered hearts, sang, built nests, and cared for their eggs.

Figure 26.16

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A phylogenetic tree of birds and their close relatives.
```

A phylogenetic tree.

The tree is as follows. Root leads to node 1. Node 1 splits to branch tip lizards and snakes and node 2. Node 2 represents common ancestor of crocodilians, dinosaurs, and birds. It splits to branch tip crocodilians and node 3. Node 3 splits to node 4 and branch tip other dinosaurs. Node 4 spits to branch tips birds and theropod dinosaurs other than birds.

(A dagger, †, indicates extinct lineages.)

VISUAL SKILLS *In this tree diagram, what is the sister taxon of the clade that includes dinosaurs and their most recent common ancestor? Explain.*

*For suggested answer, see*Appendix A.**

Internal organs, such as the heart, rarely fossilize, and it is, of course, difficult to test whether dinosaurs sang to defend territories and attract mates. However, fossil discoveries have supported the prediction that dinosaurs built nests and cared for their eggs. First, a fossil embryo of an *Oviraptor* dinosaur was found, still inside its egg. This egg was identical to those found in another fossil, one that showed an *Oviraptor* crouching over a group of eggs in a posture similar to how birds sit on their nests today **(Figure 26.17)**. Researchers suggested that the *Oviraptor* dinosaur preserved in this second fossil died while incubating or protecting its eggs. The broader conclusion that emerged from this work—that dinosaurs built nests and cared for their eggs—has since been strengthened by additional fossil discoveries that show that other species of dinosaurs built nests and sat on their eggs. Finally, by supporting predictions based on the phylogenetic hypothesis shown in Figure 26.16, fossil discoveries of nests and parental care in dinosaurs provide independent data that suggest that the hypothesis is correct.

Figure 26.17

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Fossil support for a phylogenetic prediction: Dinosaurs
built nests and cared for their eggs.
```

Fossil remains of Oviraptor and eggs. The orientation of the bones, which surround and cover the eggs, suggests that the dinosaur died while incubating or protecting its eggs.

Concept 26.4: An organism's evolutionary history is documented in its genome

As you have seen in this chapter, comparisons of nucleic acids or other molecules can be used to deduce relatedness. In some cases, such comparisons can reveal phylogenetic relationships that cannot be determined by nonmolecular methods such as comparative anatomy. For example, the analysis of molecular data helps us uncover evolutionary relationships between groups that have little common ground for morphological comparison, such as animals and fungi. And molecular methods allow us to reconstruct phylogenies among groups of present-day organisms for which the fossil record is poor or lacking entirely.

Different genes can evolve at different rates, even in the same evolutionary lineage. As a result, molecular trees can represent short or long periods of time, depending on which genes are used. For example, the DNA that codes for ribosomal RNA (rRNA) changes relatively slowly. Therefore, comparisons of DNA sequences in these genes are useful for investigating relationships between taxa that diverged hundreds of millions of years ago. Studies of rRNA sequences indicate, for instance, that fungi are more closely related to animals than to plants. In contrast, mitochondrial DNA (mtDNA) evolves relatively rapidly and can be used to explore recent evolutionary events. One research team has traced the relationships among Native American groups through their mtDNA sequences. The molecular findings corroborate other evidence that the Pima of Arizona, the Maya of Mexico, and the Yanomami of Venezuela are closely related, probably descending from the first of three waves of immigrants that crossed the Bering Land Bridge from Asia to the Americas about 15,000 years ago.

Gene Duplications and Gene Families

What do molecular data reveal about the evolutionary history of genome change? Consider gene duplication, which can play an important role in evolution because it increases the number of genes in the genome, providing more opportunities for further evolutionary changes. Molecular techniques now allow us to trace the phylogenies of gene duplications. These molecular phylogenies must account for repeated duplications that have resulted in *gene families*, groups of related genes within an organism's genome (see Figure 21.10).

Accounting for such duplications leads us to distinguish two types of homologous genes **(Figure 26.18)**: orthologous genes and paralogous genes. In **orthologous genes** from the Greek *orthos*, exact), the homology is the result of a speciation event and hence occurs between genes found in different species (see Figure 26.18a). For example, the genes that code for cytochrome c (a protein that functions in electron transport chains) in humans and dogs are orthologous. In **paralogous genes** (from the Greek *para*, in

parallel), the homology results from gene duplication; hence, multiple copies of these genes have diverged from one another within a species (see Figure 26.18b). In Concept 23.1, you encountered the example of olfactory receptor genes, which have undergone many gene duplications in vertebrates; humans have 380 functional copies of these paralogous genes, while mice have 1,200.

Figure 26.18

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Two types of homologous genes.
```

Two types of homologous genes.

Colored bands mark regions of the genes where differences in base sequences have accumulated.

Part a. Formation of orthologous genes: a product of speciation. Ancestral gene in ancestral species diverges with speciation into orthologous genes in species A and species B which differ in base sequences in parts of the gene.

Part b. Formation of paralogous genes: within a species. Ancestral gene in species C undergoes gene duplication and divergence, resulting in paralogous genes (different base sequences) in different regions on a chromosome within species C after many generations.

Note that orthologous genes can only diverge after speciation has taken place, that is, after the genes are found in separate gene pools. Thus, differences among orthologous genes reflect the history of speciation events, making them well-suited for inferring phylogeny. For example, although the cytochrome c genes in humans and dogs serve the same function, the gene's sequence in humans has diverged from that in dogs in the time since these species last shared a common ancestor. Paralogous genes, on the other hand, can diverge within a species because they are present in more than one copy in the genome. The paralogous genes that make up the olfactory receptor gene family in humans have diverged from each other during our long evolutionary history. They now specify proteins that confer sensitivity to a wide variety of molecules, ranging from food odors to sex pheromones.

Genome Evolution

Now that we can compare the entire genomes of different organisms, including our own, two patterns have emerged. First, lineages that diverged long ago often share many orthologous genes. For example, though the human and mouse lineages diverged about 65 million years ago, 99% of the genes of humans and mice are orthologous. And 50% of human genes are orthologous with those of yeast, despite 1 billion years of divergent evolution. Such commonalities explain why disparate organisms nevertheless share many

biochemical and developmental pathways. As a result of these shared pathways, the functioning of genes linked to diseases in humans can often be investigated by studying yeast and other organisms distantly related to humans.

Second, the number of genes a species has doesn't seem to increase through duplication at the same rate as perceived phenotypic complexity. Humans have only about four times as many genes as yeast, a single-celled eukaryote, even though—unlike yeast—we have a large, complex brain and a body with more than 200 different types of tissues. Evidence is emerging that many human genes are more versatile than those of yeast: A single human gene can encode multiple proteins that perform different tasks in various body tissues. Unraveling the mechanisms that cause this genomic versatility and phenotypic variation is an exciting challenge.

Concept 26.5: Molecular clocks help track evolutionary time

One goal of evolutionary biology is to understand the relationships among all organisms. It is also helpful to know when lineages diverged from one another, including those for which there is no fossil record. But how can we determine the timing of phylogenies that extend beyond the fossil record?

Molecular Clocks

We stated earlier that researchers have estimated that the common ancestor of Hawaiian silversword plants lived about 5 million years ago. How did they make this estimate? They relied on the concept of a **molecular clock**, an approach for measuring the absolute time of evolutionary change based on the observation that some genes and other regions of genomes appear to evolve at constant rates. An assumption underlying the molecular clock is that the number of nucleotide substitutions in orthologous genes is proportional to the time that has elapsed since the genes branched from their common ancestor. In the case of paralogous genes, the number of substitutions is proportional to the time since the ancestral gene was duplicated.

We can calibrate the molecular clock of a gene that has a reliable average rate of evolution by graphing the number of genetic differences—for example, nucleotide, codon, or amino acid differences—against the dates of evolutionary branch points that are known from the fossil record **(Figure 26.19)**. The average rates of

genetic change inferred from such graphs can then be used to estimate the dates of events that cannot be discerned from the fossil record, such as the origin of the silverswords discussed earlier.

Figure 26.19

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A molecular clock for mammals.
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Graph of number of mutations (vertical axis) by divergence time (millions of years, horizontal axis). Line climbs steadily from near 0 at time 0 to 75 mutations by time 120.

The number of accumulated mutations in seven well-studied proteins has increased over time in a consistent manner for most mammal species. The three green data points represent primate species, whose proteins appear to have evolved more slowly than those of other mammals. The divergence time for each data point was based on fossil evidence.

Graph of number of mutations (vertical axis) ranging from 0 to 90 with increments ever 15, by divergence time (millions of years, horizontal axis) ranging from 0 to 120 with increments every 30. The line climbs steadily from near 0 at time 0 to 75 mutations by time 120. Three data points are below the trend line, highlighted to indicate that primate proteins appear to have evolved more slowly than those of other mammals. Data table for data points given below.

INTERPRET THE DATA *Use the graph to estimate the divergence time for a mammal with a total of* 30 *mutations in the seven proteins.*

Of course, no gene marks time with complete precision. In fact, some portions of the genome appear to have evolved in irregular bursts that are not at all clocklike. And even those genes that seem to act as reliable molecular clocks are accurate only in the statistical sense of showing a fairly smooth *average* rate of change. Over time, there may still be deviations from that average rate. Furthermore, the same gene may evolve at different rates in different groups of organisms. Finally, when comparing genes that are clocklike, the rate of the clock may vary greatly from one gene to another; some genes evolve a million times faster than others.

Differences in Clock Speed

What causes such differences in the speed at which clocklike genes evolve? The answer stems from the fact that some mutations are selectively neutral—neither beneficial nor detrimental. Of course, many new mutations are harmful and are removed quickly by selection. But if most of the rest are neutral and have little or no effect on fitness, then the rate of evolution of those neutral mutations should indeed be regular, like a clock. Differences in the clock rate for different genes are related to how important a gene is. If the exact sequence of amino acids that a gene specifies is essential to survival, most of the mutational changes

will be harmful and only a few will be neutral. As a result, such genes change only slowly. But if the exact sequence of amino acids is less critical, fewer of the new mutations will be harmful and more will be neutral. Such genes change more quickly.

Potential Problems with Molecular Clocks

As we've seen, molecular clocks do not run as smoothly as would be expected if the underlying mutations were selectively neutral. Many irregularities are likely to be the result of natural selection in which certain DNA changes are favored over others. Indeed, evidence suggests that almost half the amino acid differences in proteins of two *Drosophila* species, *D. simulans* and *D. yakuba*, are not neutral but have resulted from natural selection. But because the direction of natural selection may change repeatedly over long periods of time (and hence may average out), some genes experiencing selection can nevertheless serve as approximate markers of elapsed time.

Another question arises when researchers attempt to extend molecular clocks beyond the time span documented by the fossil record. Although an abundant fossil record extends back only about 550 million years, molecular clocks have been used to date evolutionary divergences that occurred a billion or more years ago. These estimates assume that the clocks have been constant for all that time. Such estimates are highly uncertain.

In some cases, problems may be avoided by calibrating molecular clocks with data on the rates at which genes have evolved in different taxa. In other cases, problems may be avoided by using many genes rather than just using one or a few genes. By using many genes, fluctuations in evolutionary rate due to natural selection or other factors that vary over time may average out. For example, one group of researchers constructed molecular clocks of vertebrate evolution from published sequence data for 658 nuclear genes. Despite the broad period of time covered (nearly 600 million years) and the fact that natural selection probably affected some of these genes, their estimates of divergence times agreed closely with fossil-based estimates. As this example suggests, if used with care, molecular clocks can aid our understanding of evolutionary relationships.

Applying a Molecular Clock: Dating the Origin of HIV

Researchers have used a molecular clock to date the origin of HIV infection in humans. Phylogenetic analysis shows that HIV, the virus that causes AIDS, is descended from viruses that infect chimpanzees and other primates. (Most of these viruses do not cause AIDS-like diseases in their native hosts.) When did HIV jump to humans? There is no simple answer because the virus has spread to humans more than once. The

multiple origins of HIV are reflected in the variety of strains (genetic types) of the virus. HIV's genetic material is made of RNA, and like other RNA viruses, it evolves quickly.

The most widespread strain in humans is HIV-1 M. To pinpoint the earliest HIV-1 M infection, researchers compared samples of the virus from various times during the epidemic, including a sample from 1959. Comparisons of gene sequences showed that the virus has evolved in a clocklike fashion. Extrapolating backward in time using the molecular clock indicates that the HIV-1 M strain first spread to humans around 1930 **(Figure 26.20)**. A later study, which dated the origin of HIV using a more advanced molecular clock approach than that covered in this book, estimated that the HIV-1 M strain first spread to humans around 1910.

Figure 26.20

Dating the origin of HIV-1 M.

Graph of base changes between HIV sequences over time. The adjusted best-fit line fit to the cluster of data points available from 1980-2000 projects that HIV originated in the 19 thirties.

The black data points are based on DNA sequences of an HIV gene in patients' blood samples. (The dates when these individual HIV gene sequences arose are not certain because a person can harbor the virus for years before symptoms occur.) Projecting the gene's rate of change backward in time by this method suggests that the virus originated in the 1930s.

Graph of base changes between HIV sequences over time. The x-axis is year ranging from 1900 to 2000, with increments every 20 years. The y-axis is index of base changes between HIV gene sequences, ranging from 0 to 0.20 with increments every 0.05. A tight cluster of data points is shown in the top right grid of the graph, corresponding to the years 1980 to 2000 and the HIV base changes ranging from 0.10 to a little less than 0.20, with about 2/3 of the points between 0.10 and 0.15 and 1/3 of the points between 0.15 and 0.20. From this cluster of data points, an adjusted best-fit line that accounts for uncertain dates of HIV sequences is fit sloping back in time, and it crosses the x-axis in the 1930's. A range is given around the line, which ranges from about 1915 to about 1942 at the base (Y = 0), narrowing around the best-fit line as it nears the cluster of data points.

Concept 26.6: Our understanding of the tree of life continues to change based on new data

The discovery that the glass lizard in Figure 26.1 evolved from a different lineage of legless lizards than did snakes is one example of how our understanding of life's diversity is informed by systematics. Indeed, in recent decades, systematists have gained insight into even the very deepest branches of the tree of life by analyzing DNA sequence data.

From Two Kingdoms to Three Domains

Biologists once classified all known species into two kingdoms: plants and animals. Classification schemes with more than two kingdoms gained broad acceptance in the late 1960s, when many biologists recognized five kingdoms: Monera (prokaryotes), Protista (a diverse kingdom consisting mostly of unicellular organisms), Plantae, Fungi, and Animalia. This system highlighted the two fundamentally different types of cells, prokaryotic and eukaryotic, and set the prokaryotes apart from all eukaryotes by placing them in their own kingdom, Monera.

However, phylogenies based on genetic data soon revealed a problem with this system: Some prokaryotes differ as much from each other as they do from eukaryotes. Such difficulties led biologists to adopt a three-domain system. The three domains—Bacteria, Archaea, and Eukarya—are a classification level higher than the kingdom level. The validity of these domains has been supported by many studies, including a recent study that analyzed nearly 100 completely sequenced genomes.

The domain Bacteria contains most of the currently known prokaryotes, while the domain Archaea consists of a diverse group of prokaryotic organisms that inhabit a wide variety of environments. The domain Eukarya consists of all the organisms that have cells containing true nuclei. This domain includes many groups of single-celled organisms as well as multicellular plants, fungi, and animals. **Figure 26.21** represents one possible phylogenetic tree for the three domains and some of the many lineages they encompass.

Figure 26.21

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The three domains of life.
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A phylogenetic tree for the domains of life.

This phylogenetic tree is based on sequence data for rRNA and other genes. For simplicity, only some of the major branches in each domain are shown. Lineages within Eukarya that are dominated by multicellular organisms (plants, fungi, and animals) are in red type, while the two lineages denoted by an

asterisk are based on DNA from cellular organelles. All other lineages consist solely or mainly of single-celled organisms.

The tree is as follows. Root leads to node 1, common ancestor of all life. Node 1 splits to nodes 2 and 3. Node 2 splits to nodes 4 and 5. Node 4 splits to branch tip euglenozoans and nodes 6, 7, and 8. Node 6 splits to branch tip forams and node 9. Node 9 splits to branch tips diatoms and ciliates. Node 7 splits to branch tip red algae and node 10. Node 10 splits to branch tips green algae and plants. Node 8 splits to branch tip tubulinds and node 11. Node 11 splits to branch tips fungi and animals. Node 5 splits to branch tip euryarcheotes and node 12. Node 12 splits to branch tips thaumarchaeotes and crenarcheotes. Node 3 splits to nodes 13, 14, and 15. Node 13 splits to branch tips proteobacteria and mitochondria. Node 14 splits to branch tips chlamydias and spirochetes. Node 15 splits to branch tip gram-positive bacteria and node 16. Nod 16 splits to branch tips cyanobacteria and chloroplasts. Branches euglenozoans through animals represent domain Eukarya. Branches euryarcheotes through crenarcheotes represent domain Archaea. Branches proteobacteria through chloroplasts represent domain bacteria. Note: A branch point leading to multiple lineages is called a polytomy, an unresolved pattern of divergence. In this figure, one example of polytomy is the common point from which spring the various branches of the domain Eukarya.

MAKE CONNECTIONS *After reviewing endosymbiont theory (see*Figure 6.16*)*, explain the specific positions of the mitochondrion and chloroplast lineages on this tree.*

The three-domain system highlights the fact that much of the history of life has been about single-celled organisms. The two prokaryotic domains consist entirely of single-celled organisms, and even in Eukarya, only the branches labeled in red type (plants, fungi, and animals) are dominated by multicellular organisms. Of the five kingdoms previously recognized by systematists, most biologists continue to recognize Plantae, Fungi, and Animalia, but not Monera and Protista. The kingdom Monera is obsolete because it would have members in two different domains. The kingdom Protista has also crumbled because it includes members that are more closely related to plants, fungi, or animals than to other protists (see Figure 28.5).

New research continues to change our understanding of the tree of life. For example, recent metagenomic studies have uncovered the genomes of many new species of archaea, leading to discovery of the Thaumarchaeota and other previously unknown phyla of archaea. One of these newly discovered groups, the lokiarcheaota, may represent the long-sought-after sister group of the eukaryotes. If confirmed, this discovery could help shed light on how eukaryotes arose from their prokaryotic ancestors (see Concepts 27.4 and 28.1).

The Important Role of Horizontal Gene Transfer

In the phylogeny shown in Figure 26.21, the first major split in the history of life occurred when bacteria diverged from other organisms. If this tree is correct, eukaryotes and archaea are more closely related to each other than either is to bacteria.

This reconstruction of the tree of life is based in part on sequence comparisons of rRNA genes, which code for the RNA components of ribosomes. However, some other genes reveal a different set of relationships. For example, researchers have found that many of the genes that influence metabolism in yeast (a unicellular eukaryote) are more similar to genes in the domain Bacteria than they are to genes in the domain Archaea—a finding that suggests that the eukaryotes may share a more recent common ancestor with bacteria than with archaea.

What causes trees based on data from different genes to yield such different results? Comparisons of complete genomes from the three domains show that there have been substantial movements of genes between organisms in the different domains. These took place through **horizontal gene transfer**, a process in which genes are transferred from one genome to another through mechanisms such as exchange of transposable elements and plasmids, viral infection (see Concept 19.2), and fusions of organisms (as when a host and its endosymbiont become a single organism). Recent research reinforces the view that horizontal gene transfer is important. For example, one study found that on average, 80% of the genes in 181 prokaryotic genomes had moved between species at some point during the course of evolution. Because phylogenetic trees are based on the assumption that genes are passed vertically from one generation to the next, the occurrence of such horizontal transfer events helps to explain why trees built using different genes can give inconsistent results.

Horizontal gene transfer can also occur between eukaryotes. For example, over 200 cases of the horizontal transfer of transposons have been reported in eukaryotes, including humans and other primates, plants, birds, and lizards. Nuclear genes have also been transferred horizontally from one eukaryote to another. The **Scientific Skills Exercise** describes one such example, giving you the opportunity to interpret data on the transfer of a pigment gene to an aphid from another species.

Scientific Skills Exercise

Using Protein Sequence Data to Test an Evolutionary Hypothesis

AP Science Practices 2, 4, 5, 6

Did Aphids Acquire Their Ability to Make Carotenoids Through Horizontal Gene Transfer? Carotenoids are colored molecules that have diverse functions in many organisms, such as photosynthesis in plants and light detection in animals. Plants and many microorganisms can synthesize carotenoids from scratch, but animals generally cannot (they must obtain carotenoids from their diet). One exception is the pea aphid *Acyrthosiphon pisum*, a small, plant-dwelling insect whose genome includes a full set of genes for the enzymes needed to make carotenoids. Because other animals lack these genes, it is unlikely that aphids inherited them from a single-celled common ancestor shared with microorganisms and plants. So where did they come from? Evolutionary biologists hypothesized that an aphid ancestor acquired these genes by horizontal gene transfer from distantly related organisms.

Photograph of a small green aphid, an insect, on a green leaf.

How the Experiment Was Done Scientists obtained the DNA sequences for the carotenoid-biosynthesis genes from several species, including aphids, fungi, bacteria, and plants. A computer "translated" these sequences into amino acid sequences of the encoded polypeptides and aligned the amino acid sequences. This allowed the team to compare the corresponding polypeptides in the different organisms.

Data from the Experiment The sequences below show the first 60 amino acids of one polypeptide of the carotenoid-biosynthesis enzymes in the plant *Arabidopsis thaliana* (bottom) and the corresponding amino acids in five nonplant species, using the one-letter abbreviations for the amino acids (see Figure 5.14). A dash – indicates a gap inserted in a sequence to optimize its alignment with the corresponding sequence in *Arabidopsis*.

A table aligns D N A sequences for the first 60 amino acids for 6 organisms.

Data from Nancy A. Moran, Yale University. See N. A. Moran and T. Jarvik, Lateral transfer of genes from fungi underlies carotenoid production in aphids, *Science* 328:624–627 (2010).

Organism	Alignment of amino acid sequences
Acyrthosiphon (aphid)	I K I I I I I G S G Y
Ustilago (fungus)	K K V V I I G A G A
Glbberella (fungus)	K S V I V I G A G V
Staphylococcus (bacterium)	M K I A V I G A G V
Pantoea (bacterium)	K R T F V I G A G F
Arabidopsis (plant)	W D A V V I G G G H

INTERPRET THE DATA

- In the rows of data for the organisms being compared with the aphid, highlight the amino acids that are identical to the corresponding amino acids in the aphid.
- Which organism has the most amino acids in common with the aphid? Rank the partial polypeptides from the other four organisms in degree of similarity to that of the aphid.
-
    - Do these data support the hypothesis that aphids acquired the gene for this polypeptide by horizontal gene transfer? Why or why not?

- If horizontal gene transfer did occur, what type of organism is likely to have been the source?
- What additional sequence data would support your hypothesis?
- How would you account for the similarities between the aphid sequence and the sequences for the bacteria and plant?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Studies show that eukaryotes can even acquire genes by horizontal gene transfer from prokaryotes. For example, a recent genomic analysis showed that the alga *Galdieria sulphuraria***(Figure 26.22)** acquired about 5% of its nuclear genes from various bacterial and archaeal species. Unlike most eukaryotes, this alga can survive in environments that are highly acidic or extremely hot, as well as those with high concentrations of heavy metals. The researchers identified specific genes transferred from prokaryotes that have enabled *G. sulphuraria* to thrive in such extreme habitats.

Figure 26.22

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    A recipient of transferred genes: the alga *Galdieria
sulphuraria.*
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Photograph of G sulphuraria thriving on yellowish rocks near steaming hot springs. Inset shows close-up of algal cells.

Genes received from prokaryotes enable G. sulphuraria (inset) to grow in extreme environments, including on sulfur-encrusted rocks around volcanic hot springs similar to this one in Yellowstone National Park.

Overall, horizontal gene transfer has played a key role throughout the evolutionary history of life, and it continues to occur today. Some biologists have argued that horizontal gene transfer was so common that the early history of life should be represented not as a dichotomously branching tree like that in Figure 26.21, but rather as a tangled network of connected branches **(Figure 26.23)**. Although scientists continue to

debate how best to portray the earliest steps in the history of life, in recent decades there have been many exciting discoveries about evolutionary events that occurred over time. We'll explore such discoveries in the rest of this unit, beginning with Earth's earliest inhabitants, the prokaryotes.

Figure 26.23

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A tangled web of life.
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Tangled web diagram for tree of life, starting at far left with ancestral cell populations, moving through large tangled web to domains eukarya, Archaea, and bacteria on the right.

Horizontal gene transfer may have been so common in the early history of life that the base of a "tree of life" might be more accurately portrayed as a tangled web.

Tangled web diagram for tree of life, starting at far left with ancestral cell populations, moving through large tangled web to domains eukarya, archaea, and bacteria on the right. Domain Eukarya has 7 branches, including Animalia, Fungi, and Plantae (others are unlabeled). Domain Archaea, directly below Domain Eukarya, has two branches, the Euryarchaeota and the Crenarchaeota. Webbing connects these two. Domain Bacteria, below Domain Archaea, has five branches, including the Cyanobacteria and the Proteobacteria. Others are unlabeled, and webbing connects some of these branches. Approximately halfway between the ancestral cell populations (far left of the diagram) and the three domains of life (far right of the diagram), webbing labeled chloroplasts and mitochondria links the cyanobacteria (chloroplasts) and the proteobacteria (mitochondria) to the plantae (chloroplasts) and to the common ancestor of all eukaryotes (mitochondria). This webbing is shown as an arrow indicating direction of flow is from Domain Bacteria to Domain Eukarya.

Interview

Interview with Nancy Moran: Studying symbioses and horizontal gene transfer

Nancy Morgan.